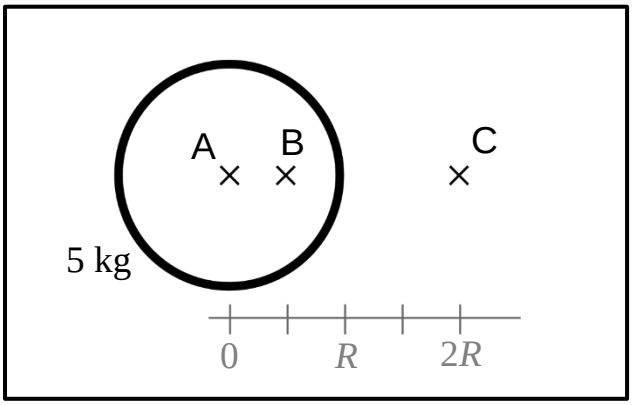
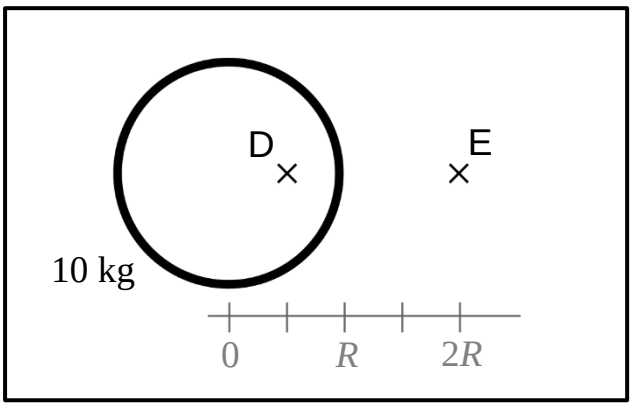
PHYS 1112 Tutorial 8

Warm-up exercise 1

In the figures below the circles represent fixed massive hollow spherical shells of the same radius R = 20 cm. Each figure is independent of the other (i.e., there is no influence whatsoever between the two).

In each situation A – E, an identical point-like object of mass 40 g is placed at the location indicated by the corresponding cross (×) symbol. The point-like object interacts with the shell through only gravity and is not under any other influences.





1. Rank the magnitude of the gravitational force experienced by the point-like object, from the largest to the smallest.

   Ranking: _____ _____ _____ _____ _____

2. Rank the gravitational potential energy of the point-like object, from the most positive to the most negative, under the usual convention that $U \to 0$ as $r \to \infty$, where r is the distance between the point-like object and the center of the shell.

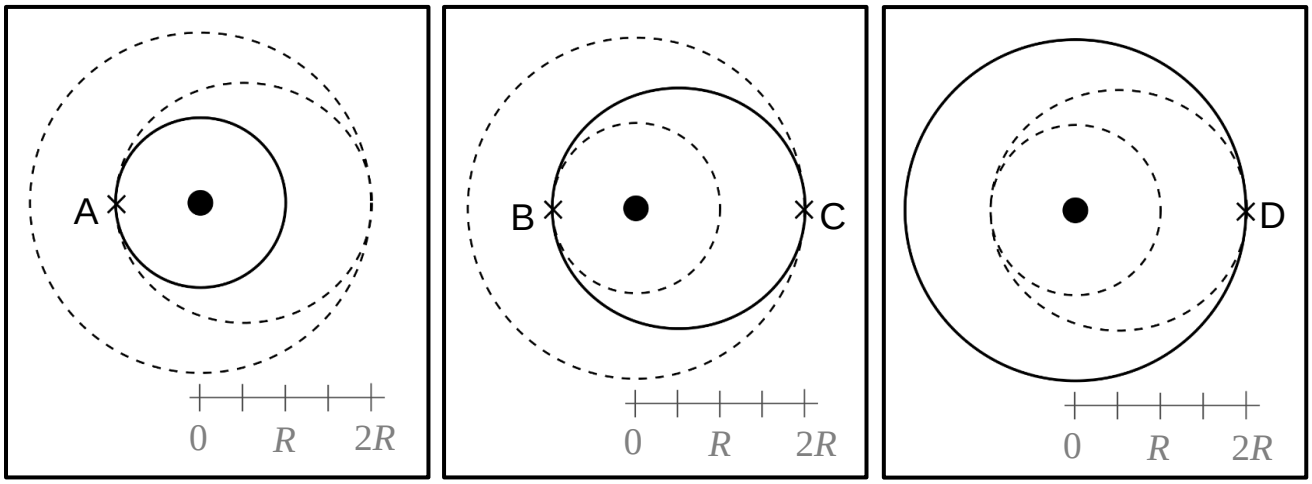
   Ranking: _____ _____ _____ _____ _____

STOP

Check with neighbors ☐ **Check with class** ☐

Warm-up exercise 2

The figures below depict an identical satellite orbiting around an identical planet. The solid line depicts the orbit that the satellite is currently on, while the crosses (×) indicate specific positions on that orbit. The broken lines illustrate the orbits from the other figures. The inner (left) and outer (right) orbits are circular while intermediate (middle) orbit is elliptical. The elliptical orbit touches the inner and outer circular orbits at its two extreme points B and C, respectively. The radiuses of the inner and outer orbits are R and $2R$, respectively.



1. Rank the situations by the speed of the satellite, from the largest to the smallest. [**Hint:** think qualitatively on the curvature of the trajectories to compare v_A and v_B (likewise for v_C and v_D)]

   Ranking: _____ _____ _____ _____

2. Rank the situations by the mechanical energy of the satellite, from the largest to the smallest.

   Ranking: _____ _____ _____ _____

3. Rank the situations by the magnitude of the angular momentum of the satellite about the center of the planet, from the largest to the smallest.

   Ranking: _____ _____ _____ _____

STOP

Check with neighbors ☐ **Check with class** ☐